

BDA syllabus.pdf

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Summary

This document outlines the **academic syllabus and laboratory plan** for a sixth-semester course in **Big Data Analytics** at BMS College of Engineering. The curriculum is structured into five units that guide students from the fundamentals of **digital data types** through the practical use of **NoSQL databases** like MongoDB and Cassandra, eventually mastering **distributed processing** with Hadoop and Apache Spark. Students are expected to achieve specific **learning outcomes** by engaging in hands-on programming tasks, such as managing **large-scale datasets** and implementing MapReduce algorithms. Evaluation is conducted through a balanced **assessment framework** consisting of internal tests, a laboratory examination, and a final comprehensive theory paper that tests various **cognitive levels** of understanding.

Keywords: Big Data Analytics, NoSQL Databases, Hadoop HDFS MapReduce, Apache Spark RDD, Scala Programming Language

Content

BMS COLLEGE OF ENGINEERING, BANGALORE-19
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Prescribed Text Books:
168
BMS COLLEGE OF ENGINEERING, BANGALORE-19
(Autonomous College under VTU Belagavi)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Reference Text Books:
E-Book:
MOOC Courses:
169
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Course Outcomes (COs):

CO-PO-PSO mapping:

Assessment Plan for CIE:

Laboratory Plan:

Instructions to Students:

1.

2.

3.

Students should write the program then start with the program execution for that week to

be executed in the lab at the beginning of each lab and get it corrected by the batch-in-charge

faculty.

Soft copy (PDF file) of all the programs along with the output needs to be submitted before

the lab test.

Continuous Internal Evaluation for each lab is for 10 marks which includes execution of the

program in the allotted lab time and showing the output. Observation book needs to be

corrected on the same day itself.

4. One Lab Test will be conducted (Mostly after theory Test-2) for all the 10 programs. Lab Test

will comprise programs which are completely different from the existing lab list of programs.

5. Marks split up for Lab:

Continuous Internal Evaluation: 100 marks → reduced to 10 marks

170

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Lab Test: 60 marks → reduced to 15 marks (Writeup- 20 marks, Execution- 30 marks, Viva- 10

marks)

Lab Programs:

171

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SEE Exam Question paper format:
172